

Five Facts about MPCs: Evidence from a Randomized Experiment

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Overview

1. A brief summary of headline results in this paper
2. A brief review of empirical MPC literature
3. Detailed results of the experiment
4. Implication for macro models & policy
5. Conclude

Introduction

Motivation: MPC is crucial for macro models and stimulus policies

- Despite **extensive** research, MPC estimates remain debated
 - Limitations arise from: Data availability, and sources of identification

Questions

1. What is the size of average MPCs, and its dynamics?
2. What observables correlate with MPCs?
3. What is the distribution of MPCs?
4. What lessons for model & policy, can we draw from empirical evidence?

Introduction

Approach is RCT: 300 euros, in a pre-paid debit card, sent in mail to ~1k individuals

- The bank constructed a representative sample of French population
- Draw treated individuals from there
- See transactions & balances in bank accounts
- Compare total consumption spending of treated HH to non-treated HH

RCT design

1. Group 1 ($N = 379$): No restrictions on transfer
2. Group 2 ($N = 268$): Any unspent expires after 3 weeks
3. Group 3 ($N = 268$): Unspent balance depreciates 10% each week
4. 50% were encouraged to spend the money quickly. Find no stat. sig. difference
5. Cannot draw cash from, or make deposit into the pre-paid card

Findings

1. G1 average MPC is 23% in first month
2. Implementation design matters: MPC much higher for G2 (61%) and G3 (35%)
3. Consumption responses concentrated on first 1-3 weeks, flat afterwards
4. What observables correlate with MPC?
 - Net liquid & illiquid wealth: Weakly negative relationship. Conditional on moderately high liquid wealth, MPC remains high
 - Average pre-treatment consumption: Negative (proxy for permanent income?)
 - Income: Negative
 - Gender: Men higher than women
 - Age: Oldest have higher MPC
5. The distribution of MPCs: A large fraction of HH have high MPCs

A BRIEF DIGRESSION: EMPIRICAL LITERATURE ON MPC

So what really is MPC? Empirics vs model

In empirics

- When people say MPC, they usually mean marginal propensity to **spend** (MPX)
 - By how much does **spending** respond to the transfer?
 - And on what category? MPX on durable & nondurable (total), MPX on nondurable
 - For durables, like cars, "consumption flow", i.e. the c in $u(c)$, accrues **slowly** over time
- Expenditure (and its dynamics) is more policy relevant

Magnitude (US 2008 tax rebates) [▶ Detail](#)

- Previous: Quarterly, total MPX 50-90%, **nondurable MPX 15-25% – target**
 - Parker et al (13), Broda and Parker (14)
- Recent estimates appear smaller: **Total MPX 25-37%, nondurable MPX 6-10%**
 - Borusyak et al (23), Orchard et al(23)

So what really is MPC? Empirics vs model

In model

- A single notional consumption summarize durable & nondurable
- Notional MPC is not the key variable in policy
 - It is something in between total MPX and nondurable MPX
- Notional MPC used for calibration
 - Laibson et al (22) gives a mapping from MPC to MPX. Roughly, $MPC = MPX / 3.2$

Definition: MPC is defined as the derivative of consumption wrt **liquid asset**

$$MPC(a, y) = \frac{c(a + x, y) - c(a, y)}{x}$$

By construction MPC will strongly correlate with liquid wealth

Question: Why is it not wrt **(transitory) income**? In reality, **policy looks more like a transitory income shock**

Empirical literature on MPC – 1

Staggered disbursement of tax rebates US: Parker et al 13, among many others

- Compare consumption response of **treated** to **not-yet treated**
- Data is CEX survey

Regular, sizable payments from Alaska Permanent Fund: Kueng (18)

- Quarterly nondurable MPC: 25%. Driven by HH with **substantial liquid asset** ($> 50\%$)
- MPC monotonically **increases in income**

Sizable lottery wins:

- US, Golosov et al (21): Average 58%, MPC decrease in income but remains high (50% for 4th income quartile)
- Norway, Fagereng et al (21): Within a year, low liquidity winners spend all, high-liquidity about 50%

Empirical literature on MPC – 2

Semi-structural: ID consumption response to transitory shock, by making assumptions on income process and consumption responses

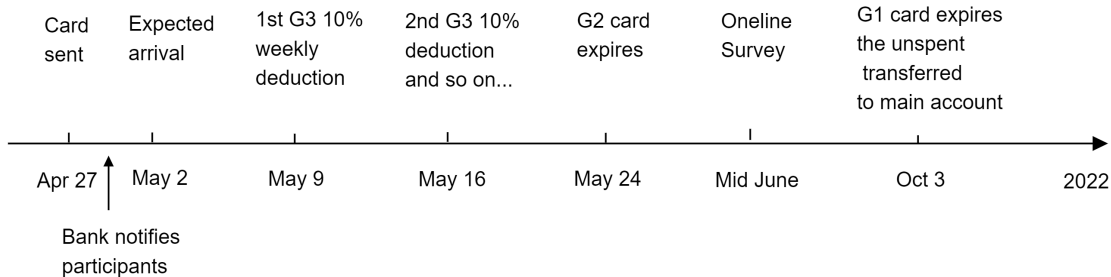
- Blundell et al (08) finds it a very small number
- Commault JMP proposes a more robust estimator. Finds a yearly MPC 32%. Reconciles with other methods

The distribution of MPC: Lewis et al (19). Use same data as Parker et al (13)

- Observables explain **very little** (8%) MPC variation. Highlight **latent heterogeneity**
- When tested jointly, only **income** (+) and **average propensity to consume** (+) (average lagged consumption divided by income) explains MPC
- Wealth data is sparse in CEX

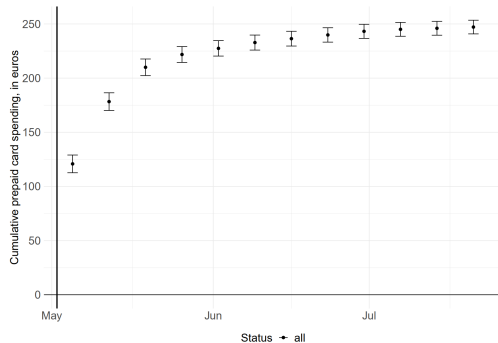
THE EXPERIMENT

Timeline

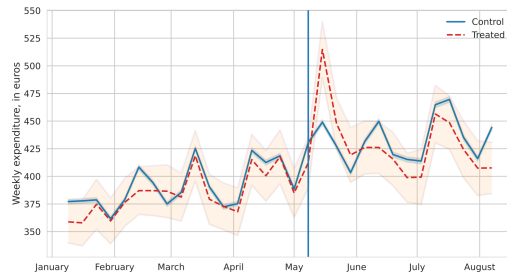


MPC in raw data

A. Cumulative Spending on Prepaid Card



B. Average Total Spending, Weekly



MPC – regression

Specification

$$Y_{it} = \sum_{\tau=0}^{\tilde{T}} \beta_{\tau} \mathbf{1}(\tau \text{ weeks since } i \text{ treated})_{it} + \alpha_i + \alpha_{tE} + \varepsilon_{it}$$

- Y_{it} : Consumption of HH i in week t , or cumulative consumption by week t .
 - Winsorized at P99 of non-treated. Robust to this.
- $\mathbf{1}(\cdot)$: HH i contains a treated individual, & week t is τ weeks after treatment
- α_i : HH FE
- E denote number of eligible individuals in a HH
- α_{tE} : Week t -by- E FE. Idea is to compare HH with same number of eligible individuals
 - For example, $\alpha_{1,2} = \mathbf{1}$ (1 week after treatment, 2 eligible individuals)

MPC – regression

My intuition for interpreting β_τ

Consider a treated HH i , a non-treated HH j , with the same E .

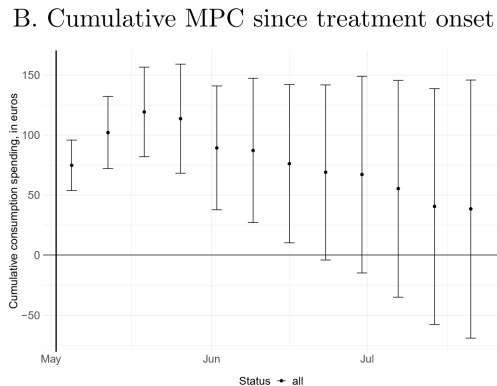
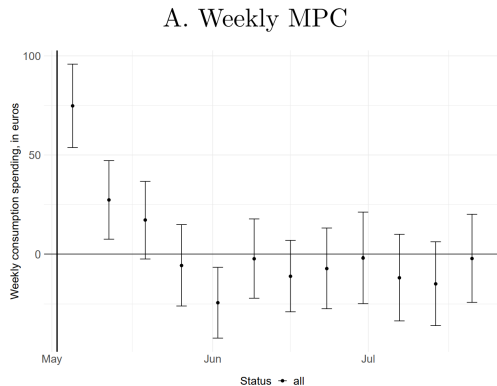
$$Y_{it} = \sum_{\tau=0}^{\tilde{T}} \beta_\tau \mathbf{1}(t - t_0 = \tau) + \alpha_i + \alpha_{tE} + \varepsilon_{it}$$

$$Y_{jt} = \sum_{\tau=0}^{\tilde{T}} \beta_\tau \times 0 + \alpha_j + \alpha_{tE} + \varepsilon_{jt}$$

Take some "average" of the difference, RCT implies last two terms are zero

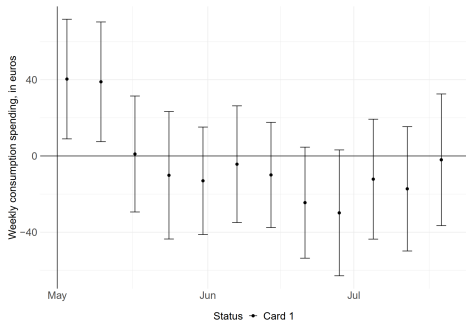
$$\begin{aligned} E(Y_{it} - Y_{jt}) &= \sum_{\tau=0}^{\tilde{T}} \beta_\tau \mathbf{1}(t - t_0 = \tau) + (E(\alpha_i) - E(\alpha_j)) + (E(\varepsilon_{it}) - E(\varepsilon_{jt})) \\ &= \sum_{\tau=0}^{\tilde{T}} \beta_\tau \mathbf{1}(t - t_0 = \tau) \end{aligned}$$

Figure 3 Pooled MPC Estimates

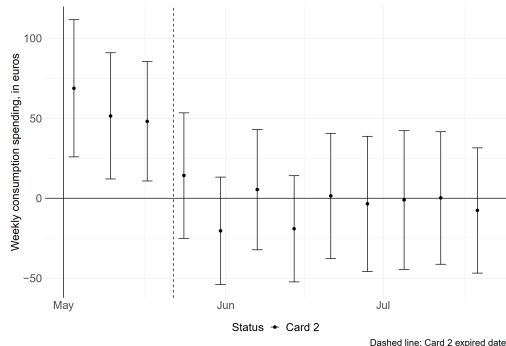


MPC - regression by card type. Y_{it} weekly consumption

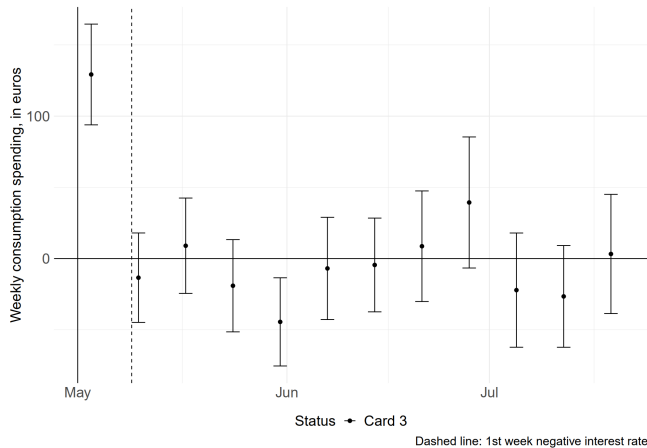
A. Group 1, no restrictions on treatment card



B. Group 2, expiration after three weeks



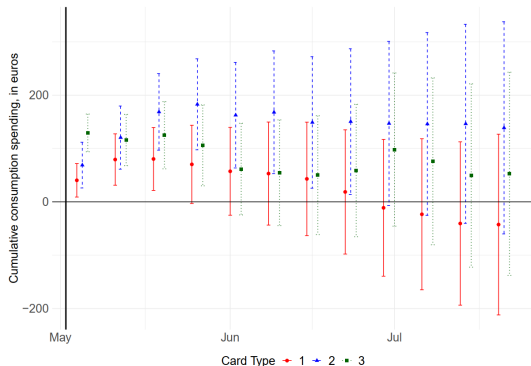
C. Group 3, negative rates every week



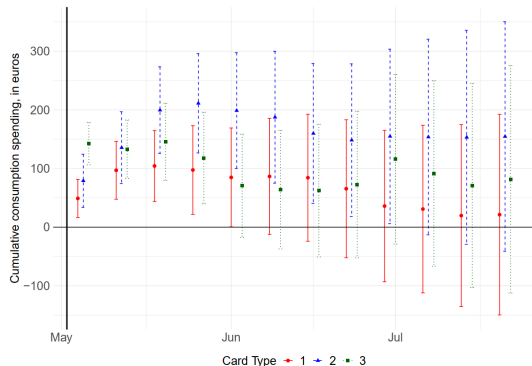
MPC - regression by card type. Y_{it} cumulative consumption

Figure 5 MPC Estimates by Treatment Group

A. Unconditional estimates



B. Conditional on take-up

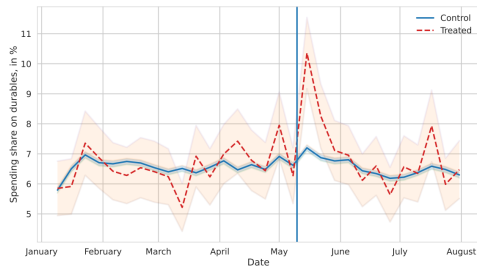


Take stock, the above establishes first 3 facts

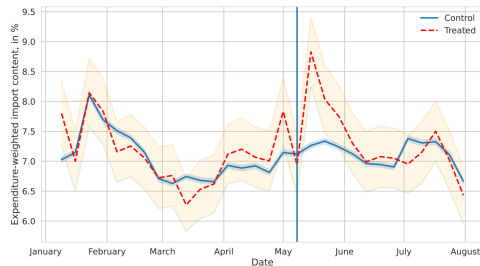
1. Average MPC in first month, on a cash like transfer, is 23%
2. Implementation design substantially affect MPC
3. Response is concentrated early on, in the first 1-3 weeks, then flat afterwards

What do people spend on?

E. Spending share on durables



F. Spending share on imported products



MPC heterogeneity by observables

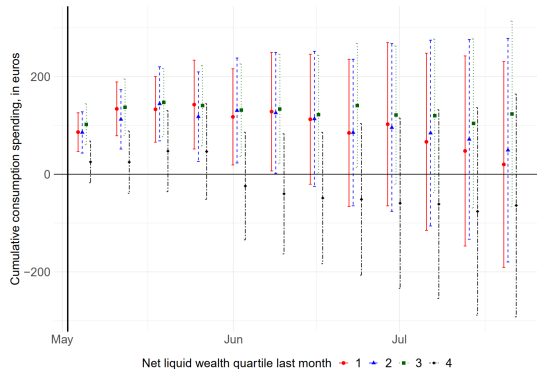
Specification

$$Y_{it} = \sum_{q=1}^4 \sum_{t=0}^{\tilde{T}} \beta_{\tau}^q \mathbf{1}(\tau \text{ weeks since } i \text{ treated})_{it} \mathbf{1}(X_i \in Q_q^X)_i + \alpha_i + \alpha_{tEQ_q^X} + \varepsilon_{it}$$

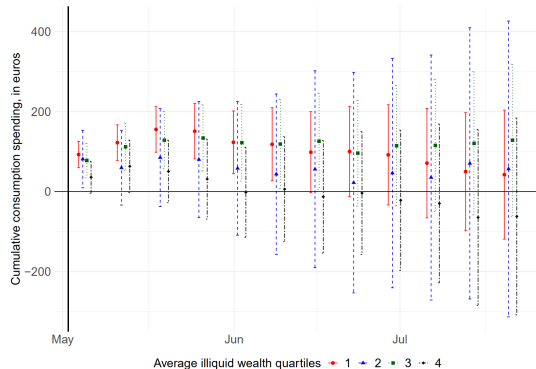
- $Q_1^X, \dots, Q_4^X$ quartiles of the distribution of time-invariant HH observable X
- Net liquid wealth: HH current account + liquid saving – short-term debt
- Net illiquid wealth: Illiquid saving + share dealing + brokerage – mortgage
- Average monthly consumption 1 year before treatment
- Income: Sum of all flow in
- Age & gender

MPC heterogeneity by observables. Y_{it} cumulative consumption

A. By Net Liquid Assets Quartile

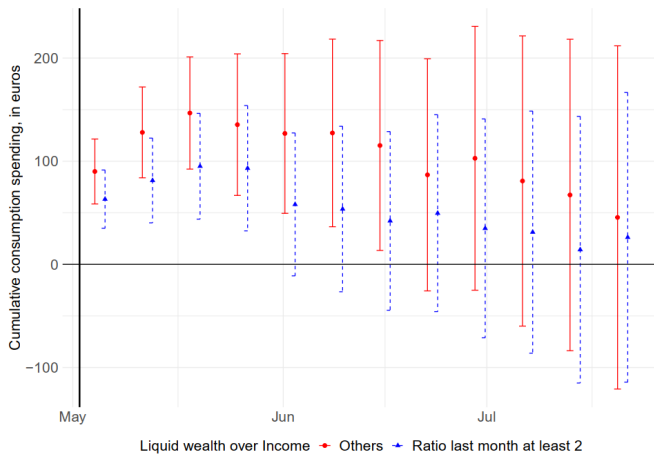


B. By Illiquid Wealth Quartile



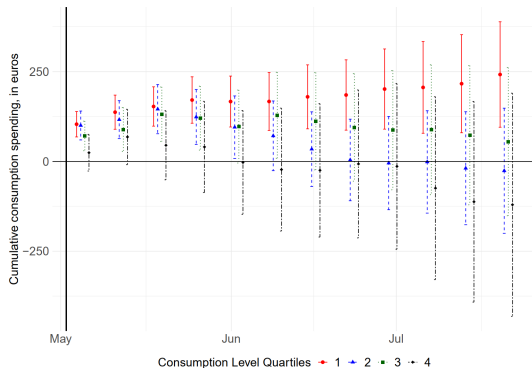
MPC heterogeneity by observables. Y_{it} cumulative consumption

A. Liquid wealth $> 2 \times$ monthly income

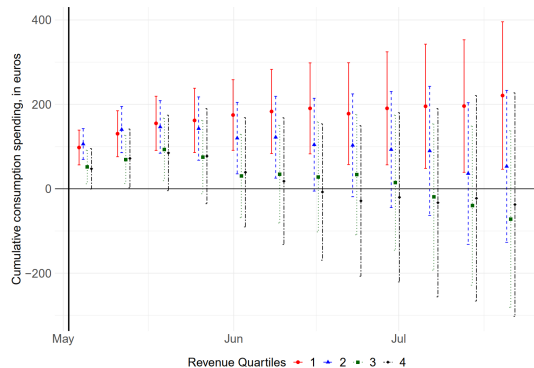


MPC heterogeneity by observables. Y_{it} cumulative consumption

C. By Average Consumption Quartile

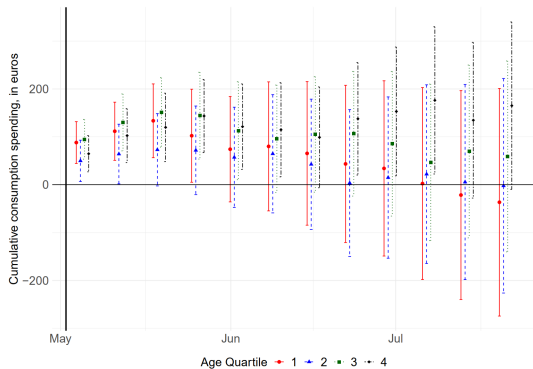


D. By Income Quartile

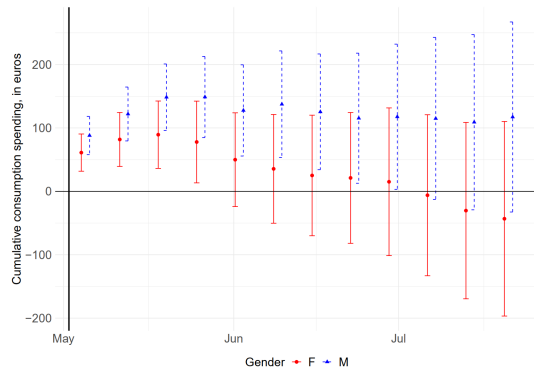


MPC heterogeneity by observables. Y_{it} cumulative consumption

E. By Age Quartile

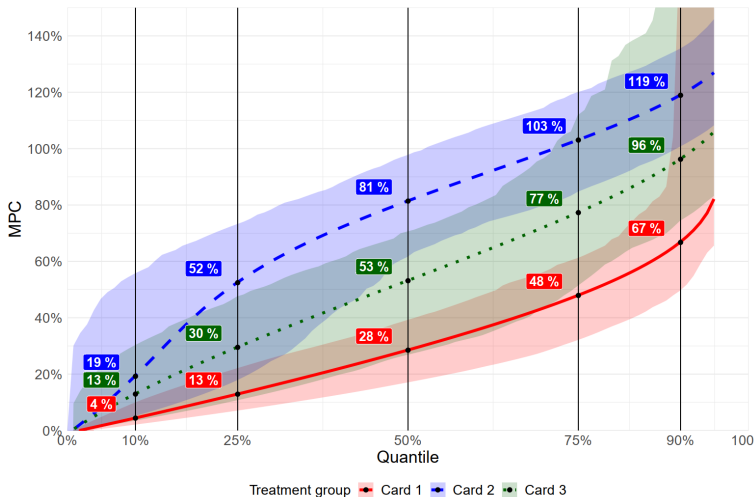


F. By Gender



Unconditional distribution of 1-month MPC $\sum_{\tau=0}^{\tilde{T}} \beta_{\tau}$

Figure 11 Household-level Quantiles of the 4-week MPC Distribution



IMPLICATIONS

Implications

Findings contrast with canonical HANK. In model

- MPC is long-lived. One-quarter 17%, two-quarter 25%, one-year 32%

This paper: No, MPC is short-lived

- MPC concentrated among a small fraction of low-liquid wealth HH

This paper: No, MPC is high for a large fraction of HH. Moreover, MPC remains high for HH with moderate or high liquid wealth

For policy

- To maximize consumption response, **implementation design** likely to matter more than targeting

Conclusion

This paper

- Fills a gap in the literature, clean identification of a small & transitory transfer
- Establishes several empirical facts inconsistent with canonical HANK
- Policy implication on design is new and important

My personal view

- Facts 1-3 seem convincing
- Is No.4 a fact? What really explains MPC?

Literature has conflicting evidence on income, vague evidence on liquid wealth

- Fact 5 points to heterogeneity in MPC, echos Lewis et al (19). But still silent on the source of heterogeneity
- Distinction between MPX and notional MPC seems important

Conclusion

This paper

- Fills a gap in the literature, clean identification of a small & transitory transfer
- Establishes several empirical facts inconsistent with canonical HANK
- Policy implication on design is new and important

My personal view

- This literature is concentrated
- If want to test model predictions, requires good data, identification, and theory
- One seemingly overlooked fact: Cross-sectional state dependence in income changes

► Detail

THANK YOU!

APPENDIX

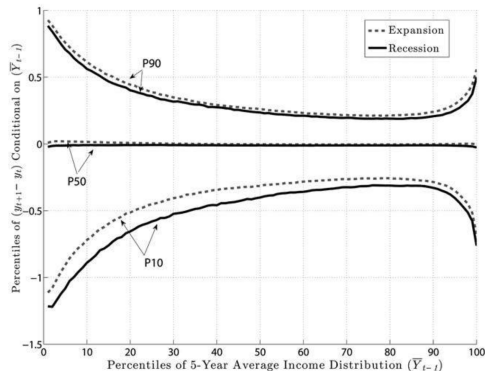
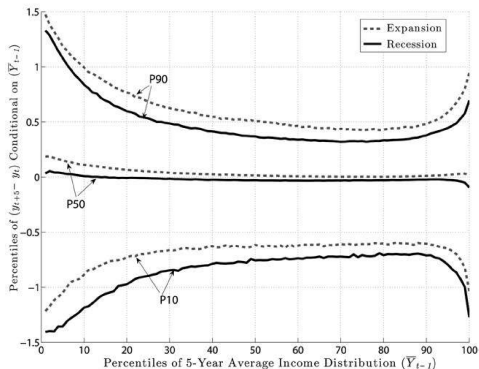
Evidence on 2008 US tax rebate

Table III First-quarter MPX and MPC Estimates for Calibration of Macroeconomic Models

	Parker et al. (2013)	Broda and Parker (2014)	Borusyak et al. (2023)	Orchard et al. (2023b)	This paper, treatment group 1
	(1)	(2)	(3)	(4)	(5)
Total MPX	52.3% to 91.1%	50.8% to 74.8%	24.8% to 36.6%	28%	23%
Nondurable MPX	12.8% to 30.8%	14.1% to 20.8%	6.9% to 10.2%	0%	6.6%
Notional MPC	16.3% to 28.5%	15.9% to 23.4%	7.8% to 11.4%	8.8%	7.2%

► Back

State dependence in income changes



Source: Guvenen et al JPE (2014) “The nature of countercyclical income risk” [▶ Back](#)